

Hazi

1.

$$f(x) := x^3 + x, \quad (x \in \mathbb{R})$$

megoldas

$$f'(x) = 3x^2 + 1 > 0 \quad (\forall x \in \mathbb{R})$$

tehát invertálható, és

$$(f^{-1})'(2) = \frac{1}{f'(1)} = \frac{1}{3 \cdot 1^2 + 1} = \frac{1}{4}$$

2.

$$f(x) := \sin\left(\frac{x-1}{x^2+1}\right) \quad (x \in \mathbb{R}), \quad a = \frac{1}{2}$$

megoldas

Mivel

$$f(a) = \sin\left(\frac{-\frac{1}{2}}{\frac{5}{4}}\right) = \sin\left(-\frac{4}{10}\right) = -\sin\left(\frac{2}{5}\right)$$

$$f'(x) = \cos\left(\frac{x-1}{x^2+1}\right) \cdot \frac{x^2+1 - (x-1)(2x)}{(x^2+1)^2} = \cos\left(\frac{x-1}{x^2+1}\right) \cdot \frac{x^2+1 - 2x^2+2x}{(x^2+1)^2} = \cos\left(\frac{x-1}{x^2+1}\right) \cdot \frac{-x^2+2x+1}{(x^2+1)^2}$$

Ekkor

$$f'\left(\frac{1}{2}\right) = -\cos\left(\frac{2}{5}\right) \cdot \frac{-\frac{4}{25}+2}{\left(\frac{4}{25}+1\right)^2} = -\cos\left(\frac{2}{5}\right) \cdot \frac{\frac{46}{25}}{\left(\frac{29}{25}\right)^2}$$

hagyjuk inkább oszinten

3.

$$f(x) := \begin{cases} 1-x, & x < 0 \\ e^{-x}, & x \geq 0 \end{cases}$$

megoldas

ha $x < 0$

$$f'(x) = -1$$

ha $x \geq 0$

$$f'(x) = e^{-x}$$

ha $x = 0$

$$\frac{f(x) - f(0)}{x - 0} = \frac{f(x) - 1}{x} = \begin{cases} \frac{1-x-1}{x} = -1 \\ \frac{e^{-x}-1}{x} \end{cases}$$

$$\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x} = \lim_{x \rightarrow 0} \frac{e^0 - e^0}{x} = ? = -1$$

Gyakorló

1/d.

$$f(x) := \frac{1}{\ln^2\left(x - \frac{1}{x}\right)} \quad (x > 1), \quad a = 2$$

$$u(x) = x - \frac{1}{x} \quad u(a) = \frac{3}{2}$$

$$v(x) = \ln(u(x)) \quad v(a) = \ln\left(\frac{3}{2}\right)$$

$$f(a) = \frac{1}{[v(a)]^2} = \frac{1}{\left[\ln\left(\frac{3}{2}\right)\right]^2}$$

$$f(x) = [v(x)]^{-2} \implies f'(x) = -2[v(x)]^{-3} \cdot v'(x)$$

$$v'(x) = \frac{1}{x - \frac{1}{x}} \cdot \left(1 + \frac{1}{x^2}\right) = \frac{1}{\frac{x^2-1}{x}} \cdot \frac{x^2+1}{x^2} = \frac{x}{x^2-1} \cdot \frac{x^2+1}{x^2} = \frac{x^2+1}{x(x^2-1)}$$

$$f'(x) = -2 \cdot \left[\ln\left(x - \frac{1}{x}\right)\right]^{-3} \cdot \frac{x^2+1}{x(x^2-1)}$$

$$f'(2) = -2 \cdot \left[\ln\left(\frac{3}{2}\right)\right]^{-3} \cdot \frac{5}{6} = -\frac{5}{3} \cdot \frac{1}{\ln\left(\frac{3}{2}\right)^3}$$

$$y = f'(2)(x-2) + f(2) = -\frac{5}{3} \cdot \frac{1}{\ln\left(\frac{3}{2}\right)^3} \cdot (x-2) + \frac{1}{\ln^2\left(\frac{3}{2}\right)}$$

1/e.

$$f(x) := x^{\ln x} \quad (x > 0) \quad a = e^2$$

$$u(x) = x \quad u'(x) = 1 \quad u(a) = e^2$$

$$v(x) = \ln x \quad v'(x) = \frac{1}{x} \quad v(a) = \ln e^2$$

$$f(a) = e^{2 \cdot \ln e^2} = e^4$$

$$f'(x) = e^{(\ln x)^2} \cdot [(\ln x)^2]' = f(x) \cdot 2 \ln x \cdot \frac{1}{x} = x^{\ln x} \frac{2 \ln x}{x}$$

$$f(a) = e^4$$

$$f'(a) = e^4 \frac{2 \cdot 2}{e^2} = e^4 \frac{4}{e^2} = 4e^2$$

$$y = f'(a)(x-a) + f(a) = 4e^2(x-e^2) + e^4 = 4e^2x - 3e^4$$

3/c.

$$\sqrt{x} + \sqrt{y} = 3, \quad (4, 1)$$

$$\sqrt{x} + \sqrt{y} = 3 \implies \sqrt{y} = 3 - \sqrt{x} \implies y = (3 - \sqrt{x})^2 \implies y = x - 6\sqrt{x} + 9$$

$$\begin{aligned} m &= \lim_{x \rightarrow 4} \frac{y(x) - y(4)}{x - 4} = \lim_{x \rightarrow 4} \frac{x - 6\sqrt{x} + 9 - 1}{x - 4} = \lim_{x \rightarrow 4} \frac{x - 6\sqrt{x} + 8}{x - 4} = \lim_{x \rightarrow 4} \frac{(\sqrt{x} - 4)(\sqrt{x} - 2)}{x - 4} = \lim_{x \rightarrow 4} \frac{(\sqrt{x} - 4)(\sqrt{x} - 2)}{(\sqrt{x} - 2)(\sqrt{x} + 2)} = \\ &= \lim_{x \rightarrow 4} \frac{\sqrt{x} - 4}{\sqrt{x} + 2} = -\frac{2}{4} = -\frac{1}{2} \\ y &= -\frac{1}{2}x + 3 \end{aligned}$$

6/a

$$f(x) := \begin{cases} ax + x^2, & x < 0 \\ x - x^2, & x \geq 0 \end{cases}$$

1. $x < 0$

$$f'(x) = a + 2x$$

2. $x > 0$

$$f'(x) = 1 - 2x$$

3. $x = 0$

1. folytonos

$$\left. \begin{aligned} \lim_{x \rightarrow 0-0} ax + x^2 &= 0 \\ \lim_{x \rightarrow 0+0} x - x^2 &= 0 \end{aligned} \right\} \implies f \in C\{0\}$$

2. derivált egyezes

$$f'_-(0) = a + 2 \cdot 0 = a$$

$$f'_+(0) = 1 - 2 \cdot 0 = 1$$

eredmeny

$$f'(x) = \begin{cases} 2x + 1, & x < 0 \\ 1, & x = 0 \\ 1 - 2x, & x > 0 \end{cases}$$

6/d.

$$f(x) := \begin{cases} \cos x, & x \leq 0 \\ a \sin x + x + b, & x > 0 \end{cases}$$

1. $x < 0$:

$$f'(x) = -\sin x$$

1. $x > 0$:

$$f'(x) = a \cos x + 1$$

1. $x = 0$

1. folytonosság

$$\left. \begin{array}{l} \lim_{x \rightarrow 0-0} \cos x = \cos 0 = 1 \\ \lim_{x \rightarrow 0+0} a \sin x + x + b = a \sin 0 + 0 + b = b \end{array} \right\} \Rightarrow \text{akkor folytonos ha } b = 1$$

2. jobb-/bal oldali derivált egyezés

$$\left. \begin{array}{l} f'_-(0) = -\sin 0 = 0 \\ f'_+(0) = a \cos 0 + 1 = a + 1 \end{array} \right\} \Rightarrow \text{akkor egyezik ha } a + 1 = 0 \Leftrightarrow a = -1$$

eredmény:

$$f'(x) = \begin{cases} -\sin x, & x < 0 \\ 0, & x = 0 \\ -\cos + 1, & x > 0 \end{cases}$$